Sucha. S , 2024ICSE0846

5CSE 12
8-8-2025

Day-01 Orientation and Environment Setup.

Date: 01/08/2026

My objectives for the day:

To understand pod created in a cloud environment launching Jupyterlab using GPU support for notebook - based work.

List of activities I did today:

Observed the complete pod creation workflow, Understood how CPU, memory, storage, and GPU resources are allocated to a pod before launch. Launched Jupyterlab using a GPU-enabled configuration, verified notebook access, basic interface setup, kernel-readiness

My critical observation:

GPU - backed notebook environments work efficiently only when the correct compute resources, runtime settings and Startup configuration are selected in advance.

What did I learn today:

Learned the purpose of pod creation, how resource allocation supports execution, exact process used to launch Jupyterlab with GPU access. Understood machine learning workflow.

What I would have done better:

Could have documented each setup step more Carefully and taken clearer notes on Configuration options reduced repeated trial and error.

My plan for tomorrow:

Revise today's setup process and begin Studying the fundamentals of AI and ML.

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Day 02: Machine learning Fundamentals and Accelerated Sklearn on GPU

Date: 02/06/2026

My objectives for the day:

To understand the meaning of AI, identify the scope, and study the basic concepts that form the foundation of ML.

List of activities I did today:

Studied the definition of AI, explored the difference between AI, ML, deep learning, and automation systems. Learned concepts such as data, features, models, training, testing, prediction. Reviewed common AI applications in daily life.

My critical observation:

ML depends heavily on useful data, proper model design, and repeated training. The quality of input data directly affects prediction accuracy.

What did I learn today:

Learned the fundamentals of AI and role of ML within AI. Understood how data-driven systems make predictions and support decisions.

What I would have done better:

Could have explore more real-world use cases from healthcare and finance sectors. More practical examples would have strengthened understanding

My plan for tomorrow:

Study neural networks and understand how neurons, layers, weights, and activation functions work together

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Day 03: Neural Networks from Scratch and Pytorch Basics.

Date: 03/06/2026

My objectives for the day:

To study neural networks, understand their internal structure, and learn how layered computation helps machines recognize patterns from data.

List of activities I did today:

Learned the concept of Artifical neurons. Studied input, hidden and output layers in neural networks. Understood the role of weights and biases in computation, reviewed activation functions and their importance in learning.

My critical observation:

Neural networks improve performance by adjusting weights and biases during learning. Even simple networks can identify useful patterns in data.

What did I learn today:

Learned the structure of neural networks and functions of neurons, layers, weights, and biases. Understood the purpose of activation functions.

What I would have done better:

Could have practiced more numerical examples to strengthen understanding.

My plan for tomorrow:

Study deep learning concepts and understand how deeper architecture improves feature learning. Explore practical applications and examples

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Day 04: CNNS and GPU profiling with NSight

Date: 04/06/2026

My objectives for the day:

To understand deep learning, relate it to neural networks. Also aimed to understand feature extraction methods

List of activities I did today:

Studied deep learning fundamentals and multi-layer neural network architectures.
Compared deep learning with traditional machine learning approaches.
Learned hidden layers extract increasingly complex features.
Observed the importance of large datasets

My critical observation:

Deep learning becomes more powerful as model depth increases, however, larger models require more data and computational resources.

What did I learn today:

Learned how deep learning automatically extracts patterns from data. Understood importance of hidden layers and feature learning.

What I would have done better:

Could have explored more practical use cases and model examples. Additional case studies would have improved

My plan for tomorrow:

Study the Antigravity tool and revise full-stack development fundamentals. understand software architecture concepts.

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Day 05: Google Antigravity and Foundation Quiz

Date: 05/06/2026

My objectives for the day:

To learn about the Antigravity tool, understand its purpose and revise the basic concepts involved in full-stack development

List of activities I did today:

Observed the purpose and usage of Antigravity tool.
Studied front-end, back-end, and database concepts.
Learned how user interfaces communicate with server-side applications.
Revised important concepts and integration principles

My critical observation:

Full-stack development requires coordination between interface design, server logic, and data management.

What did I learn today:

Learned the foundation of full-stack development and software integration.
Understood front-end, back-end and database roles.

What I would have done better:

Could have connected concepts with a practical project example. Additional architecture diagrams would have improved understanding.

My plan for tomorrow:

Revise all topics learned during the week. Organize notes clearly and prepare concise summaries

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Day 06: Transformer Architecture Deep drive

Date: 08/06/2026

My objectives for the day:

To understand transformer architecture, Study attention mechanism, and learn how modern AI system process sequential information efficiently

List of activities I did today:

Studied transformer architecture and self-attention mechanisms.
Learned encoder and decoder structure and their individual functions
Reviewed multi-head attention concepts and their contribution to model performance.
Studied practical NLP applications and modern AI systems.

My critical observation:

Transformers process information efficiently by analyzing relationships across all input elements simultaneously.

What did I learn today:

Learned transformer fundamentals and the role of attention mechanisms. Learned why transformers domain modern NLP systems.

What I would have done better:

Could have explored more implementation practical demonstrations. Spending extra time on attention calculations would have been useful.

My plan for tomorrow:

Study large language Models and efficient fine-tuning Techniques.

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Day 07: LLM fine-Tuning with LoRA and PEFT

Date: 09/06/2026

My objectives for the day:

To understand Large language Models and learn efficient adaptation techniques for specialized applications.

List of activities I did today:

Studied Large Language Models and their Capabilities

Learned LoRA and PEFT fine-tuning methods.

Reviewed transfer learning principles and adaptation strategies

prepared revision notes and important observations

My critical observation:

Efficient fine-tuning reduces computational requirements while maintaining strong performance. Fine-tuning enables models to perform specialized task effectively.

What did I learn today:

Learned how large language Models are adapted for specialized applications. Understood LoRA and PEFT techniques and their advantages.

What I would have done better:

Could have reviewed additional case studies and practical implementations. More hands-on experimentation would have improved confidence.

My plan for tomorrow:

Study Generative AI concepts and image generation systems.

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Day 08: Generative AI via Stable Diffusion XL

Date: 10/06/2026

My objectives for the day:

To understood Generative AI and explore image generation techniques used in creative AI applications.

List of activities I did today:

Studied Generative AI fundamentals and practical applications
Learned about Stable diffusion XL architecture and image generation workflows.
Explored text-to-image generation. - techniques

My critical observation:

prompt quality significantly influences generated content quality and accuracy. Well-designed prompts produce more meaningful and relevent outputs.

What did I learn today:

Learned diffusion-based image generation concepts and workflows. understood prompt driven content creation methods.

What I would have done better:

Could have tested more prompts and model configuration. Additional experimen-tation would have improved practical understandings.

My plan for tomorrow:

Study GPU optimization techniques and model performance improvements.
Revise Generative AI concepts and applications

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Day 09: GPU Optimization and Capstone Execution

Date: 11/06/2026

My objectives for the day:

To understand advanced GPU optimization techniques and improve AI model training efficiency.

List of activities I did today:

Studied mixed precision training methods and memory optimization strategies. Learned flash attention and tensor parallelism concepts. Reviewed GPU resource utilization techniques and performance improvements. Discussed optimi-zation approaches used in large-scale AI systems

My critical observation:

Optimization techniques improve training speed and reduce computational costs significantly. Hardware utilization plays a major role in large-scale-AI development

What did I learn today:

Learned advanced GPU optimization techniques and memory-efficient operations. Understood mixed precision training and performance improved methods

What I would have done better:

Could have performed additional benchmarking experiments and comparisions. More practical implementation would have improved understanding.

My plan for tomorrow:

prepare project presentation materials and revise major concepts. Organise notes clearly and review important topics.

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Day 10 : Capstone Presentation and closing ceremony.

Date: 12/06/2026

My objectives for the day:

To present the capstone project, review training outcomes, and successfully complete the AI and deep learning program.

List of activities I did today:

presented the Capstone project and explained its implementation process. participated in peer review discussion and technical presentations. Recieved feedback from mentors and trainers, Reviewed key concepts covered throughout the training program. Discussed project challenges, solutions and future improvements.

My critical observation:

Technical knowledge alone is not sufficient for successful project delivery. Communication and presentation skills are equally important.

What did I learn today:

Learned the importance of project presentation and teamwork. understood how feedback improves future work and professional growth.

What I would have done better:

Could have explained some technical details more clearly and confidently. Additional demonstration would have improved the presentation

My plan for tomorrow:

Continue exploring advanced AI technologies and practical applications. Apply acquired knowledge to future projects and research activities

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